



**EH GROUP**  
Engineering AG

# EH Fuel Cell Technology

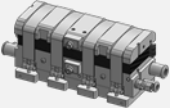
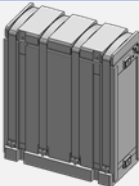
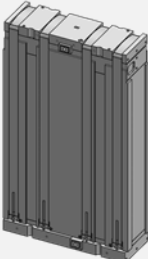
EH Group Engineering

**PEM Fuel Cell Stack & System**  
**High Power Density Stacks**  
**1.0kW – 250kW Power Range**  
**Patented Technology**

**Optimised for mobile & stationary applications**  
**Highly scalable production process**  
**Uniquely simplified architecture**  
**Swiss precision engineering**

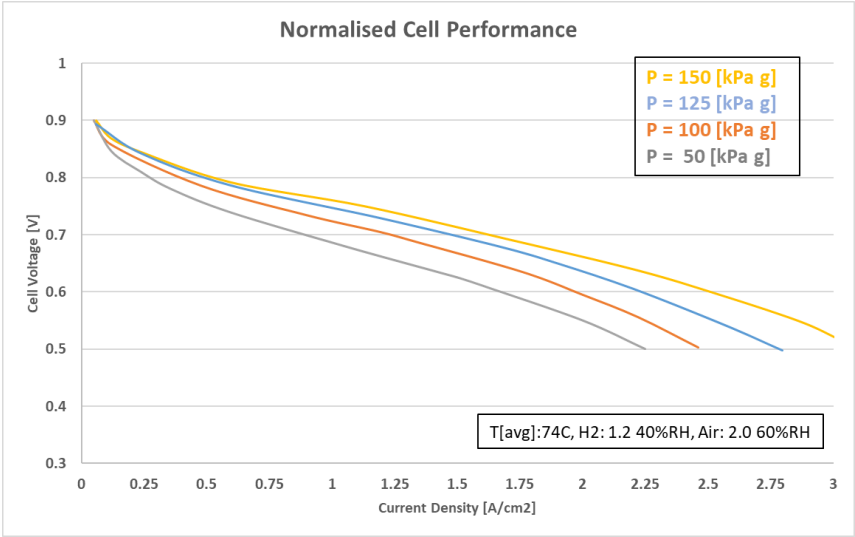
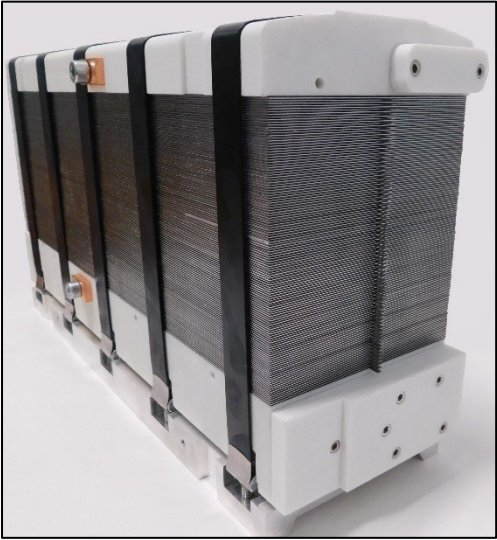


STACK PLATFORMS

| Product Name | (peak power)<br>[kW]     | Voltage<br>Range [V]                          | Current<br>Range<br>[A] | Dimensions LxWxH [mm]<br>- Stack<br>- Stack + Manifolds | Mass [kg]<br>(incl. gas manifolds) | Schematic                                                                            |
|--------------|--------------------------|-----------------------------------------------|-------------------------|---------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------|
| EH-31        | 1.1<br>2.2<br>3.2<br>4.2 | 5 – 11<br>10 – 21<br>14 – 31<br>19 – 40       | 0 – 180                 | - 200 x 75 x (12 - 50)<br>- 280 x 100 x (90 - 130)      | (4.0 - 8.0)                        |   |
| EH-51        | 5.5<br>12.0<br>16.0      | 17 – 36<br>36 – 76<br>49 – 105                | 0 – 300                 | - 280 x 140 x (45 - 146)<br>- 280 x 200 x (140 - 240)   | (10 - 19)                          |   |
| EH-81        | 22<br>37<br>68<br>104    | 36 – 95<br>62 – 160<br>112 – 294<br>172 – 450 | 0 – 450                 | - 400 x 120 x (95 - 450)<br>- 480 x 128 x (210 - 580)   | (28 - 75)                          |  |

Operating Conditions

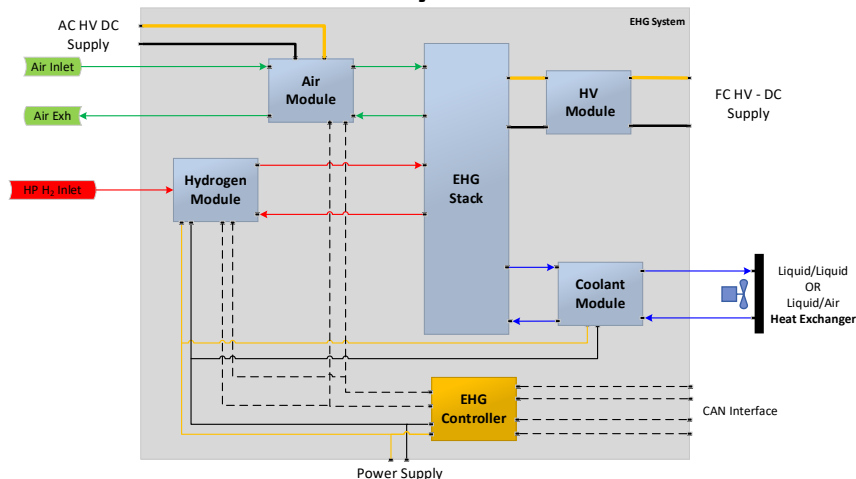
| Stream                   | Cathode                 | Anode       | Coolant                       |
|--------------------------|-------------------------|-------------|-------------------------------|
| Quality                  | Filtered, no impurities | Grade ≥ 2.5 | DI water,<br>DI + anti-freeze |
| Inlet Temp. [°C]         | 65 – 70                 | 65 – 70     | 65 – 70                       |
| Outlet Temp. [°C]        | 76 – 80 (85 short run)  | 76 – 80     | As cathode                    |
| ΔT [°C]                  |                         |             | 8 – 10                        |
| RH [%] – No condensation | 35 – 80                 | 35 – 80     | --                            |
| Stoichiometry [λ]        | 1.5 – 2.0               | 1.2 – 1.8   | --                            |



## SYSTEM PLATFORMS

| Performance                         | Power output [kW]              |                       |                       |                       |
|-------------------------------------|--------------------------------|-----------------------|-----------------------|-----------------------|
| Max power output [kW]               | 20                             | 40                    | 67                    | 106                   |
| Nominal power output [kW]           | 15                             | 35                    | 60                    | 95                    |
| Operating current range [A]         | 20 – 450 <sup>1</sup>          | 20 – 450 <sup>1</sup> | 20 – 450 <sup>1</sup> | 20 – 450 <sup>1</sup> |
| Operating Voltage range [V]         | 36 – 95                        | 62 – 160              | 112 – 294             | 172 – 450             |
| Peak system efficiency (BoL) [%]    | 60                             |                       |                       |                       |
| Nominal system efficiency (BoL) [%] | 50 – 55                        |                       |                       |                       |
| Physical                            |                                |                       |                       |                       |
| Operating temperature [°C]          | -20 – 50                       |                       |                       |                       |
| System volume [L]                   | 200 – 300 <sup>2</sup>         |                       |                       |                       |
| Environmental protection            | IP67, IP54 <sup>3</sup>        |                       |                       |                       |
| System weight [kg]                  | ~80 – 200 <sup>4</sup>         |                       |                       |                       |
| Reactants and Cooling media         |                                |                       |                       |                       |
| Cathode side (oxidant)              | Filtered, no impurities        |                       |                       |                       |
| Anode side (fuel)                   | Grade ≥ 2.5 (ISO 14687-2)      |                       |                       |                       |
| Coolant                             | DI water, DI + anti-freeze     |                       |                       |                       |
| Coolant outlet temperature [°C]     | 76-78 (85 max)                 |                       |                       |                       |
| Max system pressure [kPa g]         | 150                            |                       |                       |                       |
| Communication                       |                                |                       |                       |                       |
| Interface                           | CANbus (or other upon request) |                       |                       |                       |

### EHG Fuel Cell System Boundaries



#### Note:

- (1) Open Circuit Voltage not considered
- (2) Dimensions and packaging of system customisable upon customer requirements
- (3) Optional - upon request
- (4) May vary based on dimensions and customer requirements